

Ronak Rathod

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EDUCATION

Indian Institute of Information Technology Vadodara (IIITV)

Vadodara, Gujarat

Bachelor of Technology in Computer Science and Engineering; CPI: 7.04

2024 – 2028 (*Expected*)

- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: FastAPI, React, SQLAlchemy, NLTK, GSAP, Three.js

Databases: PostgreSQL, MySQL

Developer Tools: Git, GitHub, Docker, Render, Vercel, Vite

Concepts: REST APIs, NLP, Sentiment Analysis, Web Scraping, System Design

PROJECTS

E-Commerce Review Analyzer | *FastAPI, React, PostgreSQL, NLP, Groq AI, Docker* | [Live](#) | [GitHub](#) 2025

- Built a fake review detection ML pipeline (TF-IDF + Logistic Regression) achieving **97.5% accuracy** and **0.977 F1 score** using 5-fold stratified cross-validation on **3,000+ features** (unigram/bigram TF-IDF + 8 handcrafted features).
- Implemented DistilBERT-based sentiment classification (66M-parameter transformer) with batch processing of 32 reviews and VADER fallback, classifying reviews across 3 sentiment classes.
- Engineered aspect-based NLP analysis across **10 product dimensions** (quality, price, delivery, battery, etc.) using 110+ keywords with per-sentence sentiment scoring.
- Containerized the full-stack app with **Docker Compose (2 services)**, automating ML dependency installation, Playwright Chromium setup, and persistent data volumes for one-command deployment.
- Designed **15+ RESTful API endpoints** with rate limiting (60 req/min) and validated with **87 automated tests** (71 backend pytest + 16 frontend Vitest).
- Integrated Groq AI (**LLaMA 3.3 70B**) for structured product summaries, aspect deep-dives, and seller response suggestions with auto-retry for rate limiting.

Resume Intelligence | *Flask, Scikit-Learn, NLP, React, SQLite, Docker* | [Live](#) | [GitHub](#) 2025

- Built a **multi-signal ML scoring pipeline** combining 5 weighted factors: TF-IDF cosine similarity (n-gram 1–3), TruncatedSVD semantic embeddings, skill match precision, keyword density analysis, and structural completeness scoring.
- Engineered a **skill synonym expansion system** mapping **60+ aliases** (e.g., js→javascript, k8s→kubernetes) across **80+ technical skills** in **8 categories**, improving match recall over basic regex extraction.
- Implemented **in-memory LRU caching** (50 entries), reducing repeated analysis latency to **<1 ms**, and optimized Render deployment with Gunicorn **--preload**, **build-time NLTK pre-downloads**, and a **/health** keep-alive endpoint to eliminate cold-start delays.
- Built a React frontend with Framer Motion scroll-reveal animations, 3D tilt cards, and PDF report export.

Portfolio Website | *React, GSAP, Three.js, Vite* | [GitHub](#) 2025

- Designed and developed a premium personal portfolio with GSAP ScrollTrigger animations, glassmorphism UI, and a Three.js 3D interactive element.
- Implemented custom cursor, animated preloader, infinite tech marquee, and scroll-triggered section reveals for a polished user experience.

ACHIEVEMENTS

Data Structures & Algorithms: Solved 500+ problems on LeetCode focusing on arrays, dynamic programming, and trees using C++.

Full-Stack Deployment: Successfully deployed production applications with Docker, CI/CD pipelines, and cloud platforms (Render, Vercel).